

Electrical Machines

Engineering Technology

May, 2026

Electrical Machines (A05926)

Short Title:	Electrical Machines	
Faculty	Science and Computing	
Department	Engineering Technology	
Cluster	Electrical Engineering	
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Credits	5	Level: Introductory

Description of Module / Aims

The aim of this module is to cover the principles of electrical machine operation. Students will cover the operation and construction of Induction Motors, AC synchronous motors and generators and brushless DC motors. Motor selection and efficiency will also be covered.

Programmes

	BEng (Hons) in Electrical Engineering (WD_ETRIC_B)	2 / 4 / M
	BEng (Hons) in Electrical and Automation Engineering (International) (WD_ETRIC_BI)	3 / 5 / M
	BEng (Hons) (WD_ETRIC_B)	2 / 4 / M
ELEC-0032	BEng in Electrical Engineering (WD_EELCT_D)	2 / 4 / M
ELEC-0032	Higher Certificate in Engineering in Electrical Engineering (WD_EELCT_C)	2 / 4 / M

Indicative Content

- Principles of magnetism and magnetic circuits.
- Electromechanical energy conversion
- The construction of a synchronous machine.
- AC synchronous machines: Operation as a generator.
- Calculations on the equivalent circuit in generator mode.
- AC synchronous machines: Operation as a motor.
- Calculations on the equivalent circuit while operating as a motor.
- The construction of an induction motor.
- The operation of a three phase induction motor.
- Calculations on the equivalent circuit of an induction motor.
- Torque speed and rating plate calculations.
- The operation of Brushless DC Motors.
- Motor selection and efficiency

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Outline the principles of electrical machine operation and perform the associated analysis.
2. Describe and analyse the operation of AC synchronous machines.
3. Illustrate and explain the operation of DC and stepper motors.
4. Describe and analyse the operation of an Inductor motor.
5. Describe, analyse and utilise brushless DC motors.
6. Build, operate, assess and report on the performance of motors in laboratory setting in accordance with the module submission schedule.

Learning and Teaching Methods

- Lectures using typical classroom resources and Moodle and other on-line resources. Learning Outcome 1 to 4
- Two hour practical laboratory sessions to build, test and document. Learning Outcome 5
- Tutorials using typical classroom resources and Moodle and other on-line resources. Learning Outcome 1 to 4

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	60%	1,2,3,4,5
Continuous Assessment	40%	
<i>Practical</i>	25%	6
<i>Assignment</i>	15%	1,2,3,4,5

Assessment Criteria

70% -

100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	30	
Tutorial	6	
Practical	12	
Independent Learning	87	

Essential Material

- Hughes, D. *_Electrical and Electronic Technology_*. 9th ed.. UK: Pearson Education, 2005.
- Wildi, A. *_Electrical Machines, Drives and Power Systems_*. 6th ed.. UK: Pearson Education, 2006.

Supplementary Material

- Fitzgerald, A., B. Kingsley and C. Usmans. _Electrical Machinery_. 6th ed.. UK: McGraw Hill, ..
- Sen, P.C. _Principles of Electric Machines_. 2nd ed.. UK: Wiley, 1999.

Requested Resources

- ENGINEERING LAB: Electrical